

UNIT 3: Digital Logic — Number Systems, Gates and Sequential Circuits

Syllabus topics: Number System, Base Conversion, BCD code, Excess-3 code, Gray Code, Review of Logic Gates, Concept of Universal Gates, Boolean laws and theorems, SOP and POS representation of Boolean functions, Minimization of Boolean functions using K-map, Basic Combinational Circuits: Half Adder, Full Adder, Subtractor, Sequential Circuits: Latch, Flip-Flops, Characteristic and Excitation Table of SR, JK, D and T Flip-flop, Concept of Master Slave Flip-Flop, Shift Registers.

Question 1

Characteristic Table, Characteristic Equation				
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END SEM EXAM 2024-25 | NET-101 | CE/CSE/EE/ET/IT/ME | Q3(a) Minimal expression & realization using basic gates for $F(A,B,C,D)=\sum m(0,1,3,5,7,9,10,14,15)+d(2,4)$, Q3(b) S-R flip-flop — truth table, characteristic table, characteristic equation and excitation table

Question 2

	+ For above circuit if $v_1 = x$, $v_2 = 2x$, $v_3 = 3x$ and $R_1 = R_F/2$, $R_2 = R_F/4$ and $R_3 = R_F/2$, Then find the output v_0.		
3(a)	Explain the working of J-K Flip Flop with circuit diagram. Write its truth table. Also show Excitation table and Characteristic table.	3	3
(b)	Explain the Full Adder with its circuit diagram. Also show the expression for sum and carry output.	3	3
(c)	Solve the following using K-Map- $F(A,B,C,D) = \sum (0,3,4,7,8) + d(10,11,12,13,14,15)$ Also realize the output using logic gates	4	3
4.(a)	Show the working of LVDT with proper diagrams.	5	4
(b)	Expected value of voltage across a resistor is 90 V. However measured value is 88.5 V. Calculate (i) Percentage Error (ii) Relative Accuracy	2	4
(c)	Explain the functional block diagram of an instrument.	3	4
5.(a)	Draw the working of CRO with proper diagram. Write the measurements which can be done using CRO.	6	5
(b)	Draw the diagram of Digital Storage Oscilloscope and describe its working in	4	5

END SEM PYQ's | NET101 | CS/CE/EC/EE/IT/ME | Q3(a) J-K flip-flop working with circuit diagram, truth table, excitation & characteristic table, Q3(b) Full Adder circuit diagram + sum & carry expressions, Q3(c) K-map solve $F(A,B,C,D)=\sum(0,3,4,7,8)+d(10,11,12,13,14,15)$ & realize with logic gates

Question 3

Q2(b)	Draw the construction and working of N-Channel JFET. Draw its transfer and characteristics graph.	5	CO2	K1
Q3(a)	Find the minimal expression and realize it with basic gates. $F(A,B,C,D) = \sum m(0, 1, 3, 5, 7, 9, 10, 14, 15) + d(2, 4)$	2	CO3	K5
Q3(b)	Write down the truth table of full adder, find the minimal expression of Difference, borrow <i>sum, carry</i> and realize these expressions with basic gates.	3	CO3	K5
Q3(c)	Draw the circuit diagram of JK and T Flip Flop. Write expression for output and draw its characteristic table.	5	CO3	K4

Practice / Model Question Set | NET-101 | Q3(a) Minimal expression & realization for $F(A,B,C,D) = \sum m(0,1,3,5,7,9,10,14,15) + d(2,4)$,
Q3(b) Full adder truth table, minimal expression for sum/carry/borrow & realization, Q3(c) J-K and T flip-flop circuit diagrams,
expressions & characteristic table